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Eric W. Hoffman, Bruce E. Pinkleton, Erica Weintraub Austin & Wanda Reyes-Velázquez

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Major Article

Exploring College Students' Use of General and Alcohol-Related Social Media and Their Associations With Alcohol-Related Behaviors

Eric W. Hoffman, PhD; Bruce E. Pinkleton, PhD; Erica Weintraub Austin, PhD;
Wanda Reyes-Velázquez, PhD

Abstract. Objective: Alcohol marketers have increasingly moved their advertising efforts into digital and social media venues. As a result, the purpose of this study is to investigate associations between students' use of social media, their exposure to alcohol marketing messages through social media, and their alcohol-related beliefs and behaviors. **Participants:** Public and private university students ($N = 637$) participated November and December 2011 and April 2012. **Methods:** College students completed online surveys to measure their exposure to social and online media generally, as well as their alcohol-related digital media use and alcohol use. **Results:** Use of social media related to alcohol marketing predicted alcohol consumption and engaging in risky behaviors, whereas the use of social media more generally did not. **Conclusions:** Students' use of alcohol-related social media—marketing content associates with their problem drinking. Results have implications for alcohol abuse reduction efforts targeted at college students and suggest the importance of considering social, cultural, and cognitive factors in campaign planning and design.

Keywords: alcohol abuse prevention, alcohol-advertising effects, college students' drinking, digital alcohol marketing, social media marketing

Nationally, 80% of all college students drink, 44% binge drink, and over 65% have reported prior instances of intoxication.^{1,2} Alcohol use has been linked to violence, health problems, social problems, financial problems, unlawful behavior, and other negative con-

sequences.^{3–5} Research indicates that college-bound high school students drink less than their non-college-bound peers. This changes relatively quickly, however, as college students drink more heavily—measured as heavy-drinking occasions—than their peers not in college immediately following high school.⁶

Alcohol advertising, meanwhile, is one of a variety of factors that appear to encourage young people to drink. Young people develop their drinking-related expectancies through normative assumptions about teen drinking as well as through observations of alcohol consumption by parents, peers, and models in the mass media.⁷ Research indicates that young people with positive affective responses to alcohol advertising hold more favorable expectancies for drinking, approve of drinking at higher levels, believe drinking is more common, and have greater intentions to drink than those with less-positive responses to such advertising.⁸ Not surprisingly, longitudinal studies demonstrate that alcohol advertising, among other influences, encourages youth drinking.⁸

Alcohol marketers, meanwhile, have increasingly moved their advertising efforts into digital and social media venues, recognizing college students' near ubiquitous use of these forms of communication.^{9,10} Because researchers have yet to examine critical links among college students' social media use, exposure to alcohol-related digital marketing tactics, and problem drinking, the purpose of this study is to extend currently existing research by examining these relationships. The results of this study will help researchers better understand links among students' utilization of social media, their exposure to digital alcohol-marketing tactics, and their potentially excessive alcohol consumption.

College students' communication patterns have changed dramatically over the past several years. According to a 2011

Dr Hoffman is with the Department of Communication at Flagler College in St. Augustine, Florida. Dr Pinkleton and Dr Weintraub Austin are with the Edward R. Murrow College of Communication at Washington State University in Pullman, Washington. Dr Reyes-Velázquez is with the Communication Department at Higher Colleges of Technology in Dubai, United Arab Emirates.

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Pew Research Center study, 95% of millennials (those aged 18 to 34) own a cell telephone, 57% own a desktop computer, and 70% own a laptop computer. Moreover, 91% of this group use their mobile telephones to take a picture, 94% use their phones to send or receive text messages, 63% to access the Internet, 57% to record a video, and 57% to play a game.¹¹ A 2010 Pew study reported that 72% of online 18- to 29-year-olds use social networking sites, 33% use Twitter, and 37% share their personally created content online.¹² A content analysis of 400 randomly selected profiles of 17- to 20-year-olds on a social networking Web site revealed that 225 (56.3%) contained 341 references to alcohol, with the most frequent reference (49.3%) concerning explicit use.¹³ Clearly, it is important to understand how college students' use of digital communication tools to interact with and communicate with their peers about alcohol products and brands associates with their alcohol-related behaviors.

Researchers estimate that approximately two-thirds of alcohol marketing is conducted via new media.¹⁴ Alcohol marketing exists in a variety of Facebook features, including advertisements, pages, events, and groups; for users who list "alcohol" or "bars" as an interest, alcohol-related advertisements appear in Facebook as frequently as 1 in every 8 advertisements.¹⁰ YouTube also permits alcohol advertising for all beverage types, with many major alcohol marketers using this video-sharing site to brand and promote their products.⁹ In 2011, Brown-Forman, the manufacturer of Jack Daniels, became the first to promote alcohol on Twitter.¹⁵

Recognizing the alcohol industry's increasing reliance on social media and related forms of digital communication to marketing products, the Berkeley Media Studies Group (BMSG) issued a policy paper to alert policy makers, parents, and public health professionals of "the major features of the contemporary marketing landscape and their connection to the promotion of alcohol beverages" in 2010.^{9(p3)} The BMSG paper outlined 4 key platforms—social media, immersive environments, online video, and mobile phones—alcohol marketers use to promote their brands. Given the extent to which young people make use of social media and other forms of digital communication, and the extent to which alcohol marketing activities have migrated to digital communication environments, it is important to consider the potential influence of digital alcohol-marketing efforts on young people's alcohol consumption.

Researchers have given considerable attention to the increase in alcohol abuse taking place among young people immediately following high school. During their early 20s, widespread alcohol use among young adults progresses into pervasive use, ultimately resulting in the highest rates of average use across their adult life span, especially for college students.¹⁶ Although many individual, social, and environmental factors play a role in young people's drinking, research results indicate that associations exist between their exposure to alcohol advertising and their resulting alcohol use.⁷⁻⁹ As Aitken and colleagues point out, young people increasingly appreciate alcohol advertising with age and the act of paying attention to advertising presupposes that they

benefit or seek reward from it.¹⁷ Advertisers design their messages to entice attention and provide perceived rewards, and these advertisements appeal to and influence a broad audience, including those who are too young to drink.^{7,8,18}

In a New Zealand study, liking televised beer advertising contributed directly to alcohol consumption among 18- to 29-year-olds, whereas positive beliefs concerning the benefits of drinking contributed indirectly to alcohol consumption.¹⁹ Results from a longitudinal panel of 15- to 26-year-olds living in households in 24 US media markets indicated that participants living in markets with greater alcohol-advertising expenditures drank more than other participants, such that each additional advertisement seen by a participant increased the number of drinks consumed by 1% and each additional advertising dollar spent per capita raised the number of drinks consumed by 3%.²⁰ In fact, a review of research by Anderson and colleagues revealed that 12 of 13 longitudinal studies produced evidence indicating that exposure to alcoholic beverage advertising predicted both the onset of drinking among nondrinkers and increased levels of consumption among existing drinkers, leading the authors to conclude that alcohol advertising, among other influences, encourages drinking among young people.⁷

In terms of social media and online communication marketing, alcohol is heavily promoted on Facebook and research results indicate that heavy college drinkers are more likely to be involved in social networks on their campuses.^{10,21} Given the role of social media in students' lives, it is likely that social networking sites provide a convergence of prodrinking influences. Students potentially can be exposed to and interact with alcoholic beverage-marketing messages, publicize their drinking exploits to one another as normative, and promote opportunities to consume alcohol all via social media. Although questions remain regarding the role of social media in students' drinking, existing research indicates that social media use in general may be irrelevant to alcohol use and abuse, but engagement with alcohol marketing in social media is likely to associate with a greater probability of engaging in problem drinking behaviors. To further examine these issues, research question 1 asks if college students' use of social media related to alcohol marketing differs from their general use of social media. Research question 2 asks if college students' general use of social media associates with problem drinking behaviors. In addition, the following hypothesis is proposed based on the existing research literature.

H1: College students' use of alcohol-related social media content will associate with problem drinking.

METHODS

Researchers conducted identical online surveys at 2 universities, one a public university and the other a private university, to assess students' experiences with alcohol-related social media and online communication content. Participants at both universities were recruited from undergraduate communication courses and the surveys relied on non-probability-based, cross-sectional samples. Students

received extra credit for their participation. The Institutional Review Board (IRB) at the public university approved the study, and the IRB at the second institution accepted this approval.

Initially, 116 college students completed a pretest of potential questions to make determinations regarding scale development and to examine survey length. Following instrument refinement, the survey was conducted at a large, public university in the Pacific Northwest (61% of participants) and replicated at a small, Catholic university in the Northeast (39% of participants). Both universities are comprehensive, coeducational, and residential. Conducting the survey on both campuses provided opportunities for meaningful analytical contrasts based on campus culture. Participants' university affiliation (public or private) was included as a control in data analyses because *t* tests indicated differences existed in the responses of public university and private university students. Before beginning the survey, all participants read a briefing statement regarding survey procedures in accordance with IRB requirements. Data collection took place at both universities in November and December 2011. Additional students at the public university completed the survey in April 2012 to augment original survey participation numbers. The data from both public university surveys were combined because *t* tests indicated no differences existed in participants' responses.

Alcohol use questions in the survey were based on items used in the United States Department of Health and Human Services Substance Abuse and Mental Health Services Administration instrument,² and categorizations of alcohol-related, online communication content were based on the BMSG typology.⁹ Questions addressed participants' general social media use, their social media use related to alcohol, and their potential problem drinking behaviors. Additional questions concerned basic alcohol consumption, ownership and/or access to technology, including, for example, smart phones or tablet devices, and basic demographic questions.

Social Media Use, General

Six items composed the social media use index. Participants answered questions using 0 times, 1–4 times, 5–9 times, 10–14 times, and 15 times or more as response categories. Each question used the past 3 months as a reference point and asked participants to indicate how many times they had posted comments to an online news group, Web site, blog, or photo site ($M = 2.25$, $SD = 1.55$); created or worked on their own online journals or blogs ($M = 1.98$, $SD = 1.41$); used a social networking site such as Facebook ($M = 4.61$, $SD = 1.08$); shared something online they had created themselves, such as artwork, photos, stories, or videos ($M = 2.90$, $SD = 1.45$); used Twitter or another service to share updates about themselves or to see updates about others ($M = 3.26$, $SD = 1.79$); and posted photos or videos on social networking sites ($M = 3.16$, $SD = 1.53$). The alpha for the social media use scale was .75.

Social Media Use, Alcohol Marketing

Eight items composed the social media use index related to alcohol marketing. Participants answered questions using 0 times, 1–2 times, 3–9 times, or 10 or more times as response categories. Each question used the past 3 months as a reference point and asked participants to indicate how many times they had used Twitter or another service to receive updates about beer, wine, or liquor ($M = 1.22$, $SD = .65$); participated in contests about beer, wine, or liquor on social networking sites ($M = 1.15$, $SD = .55$); downloaded alcohol-related wallpapers/screensavers/widgets on social networking sites ($M = 1.11$, $SD = .48$); watched beer, wine, or liquor advertisements on social networking sites ($M = 1.16$, $SD = .53$); clicked on advertisements for beer, wine, or liquor on social networking sites ($M = 1.19$, $SD = .56$); used a coupon for beer, wine, or liquor found on social networking sites ($M = 1.10$, $SD = .48$); engaged in discussion in chat rooms or forums about beer, wine, or liquor ($M = 1.12$, $SD = .61$); received alcohol-related giveaways or promotional materials (such as T-shirts, key chains, or beer mugs) from a social networking site ($M = 1.33$, $SD = .67$). The alpha for the alcohol-related social media use index was .89.

Problem Drinking

Six items composed the problem-drinking index. Participants answered questions using 0 times, 1–2 times, 3–9 times, or 10 or more times as response categories. Each question asked participants if their drinking alcohol had ever caused money problems ($M = 1.31$, $SD = .69$); caused them to get in trouble at school ($M = 1.18$, $SD = .45$); hurt their schoolwork ($M = 1.49$, $SD = .77$); damaged a friendship ($M = 1.32$, $SD = .57$); made them break something ($M = 1.54$, $SD = .78$); and caused them to get hurt ($M = 1.60$, $SD = .82$). The alpha for the problem drinking index was .83.

Use Last 30 Days

Participants indicated the number of days they drank alcohol during the past 30 days using 0 days, 1–2 days, 3–5 days, 6–9 days, 10–19 days, 20–29 days, all 30 days as response categories ($M = 3.27$, $SD = 1.47$).

Use in 1 Occasion

Participants indicated the amount of alcohol they usually drink in 1 occasion. Response categories included I don't drink, a single drink, enough to feel it a little, enough to feel it a lot, and until I get really drunk ($M = 3.10$, $SD = 1.11$).

RESULTS

The combined surveys produced 737 usable questionnaires. Of participants reporting demographic characteristics, the sample included 441 (68%) females and 205 (32%) males. The average age of all respondents was 21.4 years old. Survey participants could indicate more than 1 race or ethnicity and of those who responded, most reported being Caucasian ($n = 561$, 76%). Approximately 3% ($n =$

22) of the sample reported being African American, 7% ($n = 49$) reported being Asian, and 4% ($n = 32$) reported being Hispanic. In addition, 3% ($n = 19$) reported other and 7% declined to report their ethnicity. In many ways, the samples were generally representative of the populations from which they were drawn, but overrepresented females and slightly underrepresented minorities at the public university, for example.

Approximately 96% ($n = 613$) of sample members reporting their media consumption used social media, with 80% ($n = 519$) posting photos or videos on social networking sites and 68% ($n = 445$) using Twitter or a similar microblogging service. Fewer than half of participants reporting their online media use patterns (43%, $n = 275$) reported using an online journal or blog, whereas 78% ($n = 510$) of respondents reported they had shared something they created themselves online (eg, photos or videos) and 50% ($n = 327$) reported posting a comment to an online news group, Web site, blog, or photo site.

In terms of alcohol-related online media use, 50% ($n = 329$) of participants reported that they had watched an alcohol-related video on a social media site, 38% ($n = 249$) visited an alcoholic beverage Web site, and 35% ($n = 281$) indicated watching advertisements for alcoholic beverages on social networking sites. Meanwhile, 14% ($n = 90$) reported clicking on alcoholic beverage advertisements and 15% ($n = 101$) said they used Twitter or another micro blogging service to receive updates about alcohol. In addition, 19% ($n = 114$) indicated that they had purchased an alcohol-related product through social media. A total of 12% ($n = 78$) of respondents engaged in chat room discussions or forums about alcohol, whereas 11% ($n = 69$) said they participated in contests about alcohol on social networking sites. Only 7% ($n = 47$) of respondents reported downloading alcohol-related wallpapers/screensavers/widgets on social networking sites and 6% ($n = 38$) of respondents used a coupon for alcohol found on social networking sites in the 3 months prior to the survey.

To answer research question 1 concerning the distinctions between students' use of alcohol-related online and social media-marketing content and their general use of online and social media, the authors performed a principal components factor analysis. Two factors emerged, explaining 47% of the variance. The 2 factors included a general use of social media factor, which consisted of 6 items and explained 33% of the variance, and a social media use factor related to alcohol marketing, which consisted of 8 items and explained an additional 14% of the variance. The factor correlation was $r = .16$.

Two composite scales based on the 2 factors demonstrated acceptable reliability, with an alpha = .74 for the social media use index, $M = 3.04$, $SD = .97$, and an alpha = .89 for the social media use related to alcohol marketing index, $M = 1.19$, $SD = .45$. The 2 indices were modestly correlated, $r = .15$, $p < .001$. These results confirmed that use of social media involving alcohol marketing is distinct from the general use of social media, in answer to research question 1.

Additional analyses revealed the public university students used more social media ($M = 3.11$) than the private, religious university students ($M = 2.92$), $t(545) = 2.16$, $p < .05$, and the public university students also used more social media related to alcohol marketing ($M = 1.23$) than the private university students ($M = 1.12$), $t(545) = 3.09$, $p < .01$.

Research question 2 concerned students' use of social media generally and its relationship to problematic drinking, whereas hypothesis 1 concerned students' use of social media alcohol-marketing applications and its relationship to problematic drinking. For purposes of analysis, the authors used a hierarchical multiple regression analysis with demographic control variables entered in the first block, followed by the social media indices in the second block. Demographic variables included sex, age, reported family income, reported grades in school, expectations for educational attainment, year in college, and university affiliation. Three dependent variables were tested. For ease of interpretation, standardized coefficients for variables other than categorical variables are reported, for which unstandardized betas are reported. To account for the potential role of school culture, school was entered as a control in the analyses. In addition, the analysis tested for possible interaction effects of school with other variables, and no interactions existed.

For the dependent variable of demonstrated drinking problems ($\alpha = .85$), demographic predictors explained 13% of the variance, $F(7, 601) = 12.72$, $p < .001$, as shown in Table 1. The social media variables explained an additional 7% of the variance, $F(9, 599) = 17.67$, $p < .001$. Worse grades in school predicted more drinking problems, $\beta = -.13$, $p < .01$, as did having been in college longer, $\beta = .10$, $p < .05$, whereas attending a private, religious university predicted fewer drinking problems, $\beta = -.19$, $p < .001$. Other demographic variables were not significant predictors. In addition, the use of social media alcohol-marketing applications predicted more drinking problems, $\beta = .28$, $p < .001$, in support of hypothesis 1. The use of social media more generally was not a significant predictor, in answer to research question 2.

For the dependent variable of alcohol use frequency during the past 30 days, demographic predictors explained 25% of the variance, $F(7, 601) = 29.30$, $p < .001$. The social media variables explained an additional 6% of the variance, $F(9, 597) = 30.22$, $p < .001$. Higher family income predicted more frequent consumption, $\beta = .12$, $p < .001$, as did sex (male), $\beta = -.08$, $p < .05$, more years in school, $\beta = .21$, $p < .001$, and lower grades, $\beta = -.07$, $p = .05$. Attendance at a private, religious university predicted less frequent consumption, $\beta = -.30$, $p < .001$. Other demographic variables were not significant predictors. In addition, the use of social media alcohol-marketing applications predicted more frequent consumption, $\beta = .24$, $p < .001$, in support of hypothesis 1. The use of social media more generally was not a significant predictor, in answer to research question 2.

For the dependent variable of consumption on a single occasion, demographic predictors explained 17% of the variance, $F(7, 602) = 17.14$, $p < .001$, and social media variables explained an additional 6% of the variance, $F(9, 600)$

TABLE 1. Regressions Predicting Alcohol Use Behaviors Based on Demographic Characteristics and Social Media Use

Dependent variable	Independent variables	β	R^2	$F(df)$
Problem drinking behavior index	Block 1		.13	12.72(7, 601)***
	Family income	.00		
	Sex	-.08*		
	Year in school	.10*		
	Academic achievement	-.13*		
	Age	-.02		
	School type	-.19***		
	Academic expectations	-.02		
Alcohol use frequency (past 30 days)	Block 2		.07	17.67(9, 599)
	General social media use	.03		
	Alcohol-related social media use	.28***		
	Block 1		.25	29.30(7, 601)***
	Family income	.12***		
	Sex	-.08*		
	Year in school	.21***		
	Academic achievement	-.07 [†]		
Usual consumption on a single occasion	Age	-.04		
	School type	-.30***		
	Academic expectations	-.02		
	Block 2		.05	30.22(9, 599)***
	General social media use	.03		
	Alcohol-related social media use	.24**		
	Block 1		.17	17.14(7, 602)***
	Family income	.08*		
	Sex	.00		
	Year in school	.14**		
	Academic achievement	-.12**		
	Age	-.04		
	School type	-.28***		
	Academic expectations	.06		
	Block 2		.05	19.01(9, 600)***
	General social media use	.05		
	Alcohol-related social media use	.23***		

Note. Sex is a dichotomous variable (higher value denotes female). School type is a dichotomous variable (higher value denotes private religious institution). For other variables, a higher score denotes higher levels, more positive attitudes, or greater frequency.

[†] $p = .05$; * $p < .05$; ** $p < .01$; *** $p < .001$.

= 18.18, $p < .001$. Worse grades in school predicted more consumption, $\beta = -.12$, $p < .01$, as did more years in school, $\beta = .14$, $p < .01$, and higher family income, $\beta = .08$, $p < .05$. Attendance at a private, religious university predicted less consumption, $\beta = -.28$, $p < .001$. The use of alcohol marketing in social media predicted more consumption in a single occasion, $\beta = .23$, $p < .001$, in support of hypothesis 1. The use of social media generally was not a significant predictor, in answer to research question 2.

In sum, the results showed that participants' use of digital media alcohol-marketing applications predicted their alcohol consumption and problem drinking. In contrast, students' general social media use did not predict their alcohol consumption or problem behavior.

COMMENT

College students' communication practices have changed dramatically over the past several years, in ways that parallel

changes made by alcohol marketers. Given that marketers' alcohol sales programs have shifted to online platforms just as students' communication patterns have shifted to online platforms, it is important to consider the potential implications of these changing patterns for potential alcohol-marketing uses and effects on this population. As a result, the purpose of this study was to examine college students' use of social media generally and their experiences with alcohol-marketing applications in social media specifically, to determine how these experiences associate with alcohol consumption and related problematic behaviors such as having financial problems, getting in trouble at school, having difficulties academically and socially, and damaging something or getting hurt.

Study results indicate that students' use of social media alcohol-marketing applications is distinct from their general social media use and that their use of social media related to alcohol marketing associates with higher levels of consumption and problem behaviors attributable to drinking. Although

the results of this cross-sectional study do not provide for a determination of causation, they do provide evidence suggesting interaction with online alcohol-marketing messages and that related social media contents have the potential to contribute to abusive drinking among college students.

Existing research indicates that alcohol-advertising messages in traditional media often contain appeals that young people and college students find attractive and, as a result, influence their decisions about drinking. These include, for example, appeals containing humor, animation, sexual allure, romance, escape, wealth, and adventure, and these messages often rely on slice-of-life depictions that focus on drinkers and drinking occasions rather than on product qualities.^{8,10,22} The result, according to researchers, is that exposure to alcoholic beverage advertising in traditional media contributes to earlier initiation of drinking, higher levels of consumption, and positive expectancies regarding the perceived benefits of drinking.^{7,20} Given advertisers' ability to microtarget potential consumers using social media and to immerse them in online marketing environments such as games and contests,⁹ alcohol-marketing content in social media has the potential to pose even greater risks for promoting alcohol abuse than traditional marketing.

Together with existing research, the results of this study suggest that alcohol-marketing programs using social media have the potential to increase consumer–brand engagement among college students. Researchers generally recognize that personal relevance and the enhanced importance of a topic to an individual are associated with an individual's enduring involvement level—defined as long-term, inherent interest in a topic or product category.^{23,24} The consequences of involvement may include increased purchase behavior and greater perceived product attribute differences among other outcomes.^{25,26} Other research findings indicate that individuals pay greater attention to personally relevant messages and that when individuals are high in involvement, involvement-relevant messages attract more attention and cognitive effort than other messages.^{26–29}

In addition, because heavy drinkers tend to be more involved in social networks,²¹ the potential exists for the effects of online alcohol marketing to be especially magnified among this subgroup. Research indicates that the sharing of beliefs, attitudes, and behaviors among young adults often includes references and depictions of alcohol use that facilitate the development and reinforcement of social norms, including those regarding dangerous alcohol consumption and related behaviors.^{30–32} This presents a particularly acute concern because many college students are underage drinkers and make poor decisions regarding alcohol consumption.^{1,2,6} Decision making under the influence of alcohol can lead to impulsive social media interactions that can promote or put students at increased risk of dangerous alcohol-related behaviors.^{3–5} From a marketing perspective, social media can help cultivate brand identification and loyalty among potential new customers, with a quite large potential payoff over the course of numerous purchase decisions and years of consumption. The ability of social media to contribute to consumer–brand

interactions and to create and strengthen brand relationships is a key benefit of digital marketing messages.^{33,34}

In this study, students' use of social media to participate in alcohol marketing explained between 6% and 10% of the variance in problematic behaviors associated with alcohol use, frequency of consumption, and amount consumed on a single occasion. This relationship emerged even after demographic variables representing the cultural context had been entered into the regression analyses. From a campus public health perspective, these results suggest that social media use in general does not pose a risk factor but can be part of a constellation of tools used by alcohol marketers to reach potential consumers with their messages. Given that 96% of students were engaged in social media use, these results indicate that social media can provide a powerful and likely essential tool for health promotion campaigns regarding the dangers of alcohol use. Just as marketing campaigns can create popular viral content, so can social marketing campaigns if the messages are well produced and targeted effectively.^{35–37}

These results of this study likely represent a reciprocal relationship between students' use of alcohol-marketing content in social media and alcohol-related behaviors rather than an exclusively predictive one. For example, those interested in alcohol use can be expected to seek out alcohol-marketing messages more frequently than those less interested. This interpretation of the results is supported by the fact that the private, religious school students, who tended to consume alcohol somewhat less frequently than the public university students, also tended to use social media for alcohol marketing less frequently, with the use of social media for marketing explaining variance in drinking frequency beyond that explained by school affiliation. In other words, the results of this study suggest that the cultural context may contribute to students' interest in drinking alcohol, spurring relevant information seeking that facilitates use.²¹ Prevention specialists should examine these variables further in future research and consider audience members' perspectives, cultural context, and related information in prevention-campaign programming.²²

This interpretation of the results is further supported by the finding that social media variables primarily represent user-initiated behaviors rather than passive receipt of messages. In most cases, the media user has to either seek out or respond actively to participate in the alcohol-marketing–related content. For example, students must subscribe to receive updates about beer, wine, or liquor, enter a contest to participate, or must post a comment in order to engage in an online discussion. Even simply clicking on an advertisement or a link posted by someone else represents active engagement with the content. As a result, researchers should examine the role of specific forms of social media that associate with abuse and examine users' motivations and information source use as they relate to and promote alcohol consumption.

Given the importance of information-processing and message interpretation processes in media effects related to alcohol marketing,^{17,18} the associations documented here

between alcohol use and social media for alcohol marketing probably are greater than indicated in this study. A number of studies have shown that individuals' receptivity to alcohol-marketing messages is a more powerful predictor than exposure, per se.^{17,18,37} As a result, media literacy education, where young people learn to thoughtfully examine and deconstruct media messages, may be an important tool to include in health promotion campaigns. Some research has suggested, for example, that simply increasing media users' awareness of the persuasive intent of marketers can increase their skepticism, and reduce the influence of appealing messages.^{38–40} Given this, a media literacy-based approach to alcohol abuse prevention campaigns may have potential to help reduce the influence of social media alcohol-marketing content on youth and college students and researchers should examine this possibility in future research.

Limitations

This study has limitations that necessitate careful interpretation of its results. Initially, the samples are non-probability based and do not perfectly represent their populations. Next, the methods used by researchers do not provide for a determination of causation. Finally, this study only examined participants' usage of social media and social media alcohol-marketing content and did not examine their message interpretation processes. Accounting for respondents' message interpretation processes is likely to provide a more accurate and complete examination of participants' interaction with alcohol marketing, and researchers should consider individuals' message interpretation processes in future research.

Conclusions

The results of this study demonstrate that college students' use of social media presents both challenges and opportunities for health promotion advocates. The confluence of students' shift to social media as a preferred communication tool along with marketers' similar shift increases the probability that students interacting with alcohol-marketing content in social media will reinforce and exacerbate risky alcohol use behaviors and their consequences. Social media use itself does not appear to increase students' risk of alcohol use and related problems, and, as a result, prevention campaigns also should make use of these message outlets. This means that campus health promotion programs can facilitate the use of social media by students as advocates for one another, potentially reinforcing positive norms and promoting healthy behaviors. It also means that health practitioners may find it useful to include media literacy education among their campaign strategies to increase students' awareness of media influences and their skepticism toward alcohol-marketing strategies.

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CONFLICT OF INTEREST DISCLOSURE

The authors have no conflicts of interest to report. The authors confirm that the research presented in this article met the ethical guidelines, including adherence to the legal requirements, of the United States and received approval from the Institutional Review Board of Washington State University.

NOTE

For comments and further information, address correspondence to Eric W. Hoffman, 74 King Street Street, Augustine, FL 32084, USA (e-mail: ehoffman@flagler.edu).

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